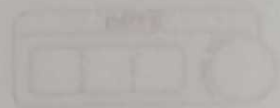


18/13/26



Q1] Evaluate $\int_0^{\infty} \frac{x^3}{3^x} dx$

$$\rightarrow I = \int_0^{\infty} \frac{x^3}{3^x} dx = \int_0^{\infty} x^3 e^{-x \ln 3} dx$$

use gamma integral

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$n=3; a=\ln 3$$

$$I = \frac{3!}{(\ln 3)^4} = \frac{6}{(\ln 3)^4}$$

Q2] Evaluate $\int_0^{\infty} x^n e^{-x^m} dx$

$$\text{Let } t = x^m \Rightarrow x = t^{1/m}$$

$$dx = \frac{1}{m} t^{1/m-1} dt$$

$$I = \int_0^{\infty} t^{n/m} e^{-t} \frac{1}{m} t^{1/m-1} dt$$

$$= \frac{1}{m} \int_0^{\infty} t^{\frac{n+1}{m}-1} e^{-t} dt$$

using gamma function,

$$\Gamma(k) = \int_0^{\infty} t^{k-1} e^{-t} dt$$

$$I = \frac{1}{m} \Gamma\left(\frac{n+1}{m}\right)$$

Q3] Express $\int_{-1}^1 (1+x)^m (1-x)^n dx$ in terms of gamma function

$$I = \int_{-1}^1 (1+x)^m (1-x)^n dx$$

$$\text{put } x=1-2t; dx=-2dt$$

$$x = -1 \Rightarrow t = 1, \quad x = 1 \Rightarrow t = 0$$

$$I = 2^{m+n+1} \int_0^1 t^n (1-t)^m dt$$

using beta function,

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt = \frac{\Gamma(p) \Gamma(q)}{\Gamma(p+q)}$$

$$I = 2^{m+n+1} \frac{\Gamma(m+1) \Gamma(n+1)}{\Gamma(m+n+2)}$$

Q4] show that $\int_0^{\infty} \frac{x^{n-1}}{(a+bx)^{m+n}} dx = \frac{1}{a^m b^n}$

$$I = \int_0^{\infty} \frac{x^{m-1}}{(a+bx)^{m+n}} dx$$

putting $x = \frac{a}{b} t, \quad dx = \frac{a}{b} dt$

$$I = \frac{a^{-n}}{b^m} \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} dt$$

using beta function identity

$$\int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} dt = B(m, n)$$

$$I = \frac{1}{a^n b^m} B(m, n)$$

hence proved

Q5) Prove that $\int_0^{\infty} \frac{e^{-ax} \sin x}{x} dx = \cot^{-1} a$

$$I(a) = \int_0^{\infty} \frac{e^{-ax} \sin x}{x} dx \quad (I)$$

$$I'(a) = \frac{d}{da} \int_0^{\infty} \frac{e^{-ax} \sin x}{x} dx$$

$$= \int_0^{\infty} \frac{\partial}{\partial a} \frac{e^{-ax} \sin x}{x} dx$$

$$= \int_0^{\infty} \frac{-x e^{-ax} \sin x}{x} dx$$

↳ standard result

$$= - \int_0^{\infty} e^{-ax} \sin x dx$$

$$= \frac{-e^{-ax} [a \sin bx - b \cos bx]}{(a)^2 + b^2}$$

$$= - \left[\frac{e^{-ax}}{a^2 + 1} \right] (-a \sin x - \cos x) \Big|_0^{\infty}$$

$$= - \left[\frac{1}{a^2 + 1} \right] (-1)$$

$$I'(a) = \frac{-1}{a^2 + 1}$$

$$\frac{dI}{da} = \frac{-1}{a^2 + 1}$$

$$dI = \int \frac{-1}{a^2 + x^2} dx$$

$$= -\tan^{-1}\left(\frac{x}{a}\right) + C \quad \text{--- (II)}$$

Put (i) at the place of 'a' the equation so it will become

$$\int_0^{\infty} \frac{e^{-ax} \sin x}{x} dx = 0$$

put the value in (II)

$$= -\tan^{-1}\left(\frac{\infty}{a}\right) + C$$

$$= -\frac{\pi}{2} + C$$

$\left[C = \frac{\pi}{2}\right]$ so equation 2 becomes

$$I(a) = \frac{\pi}{2} - \tan^{-1}\left(\frac{x}{a}\right)$$

$$I(a) = \cot^{-1}\left(\frac{x}{a}\right)$$

hence proved

$$\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2} \quad \text{this can possible if } a=0$$

$$\int_0^{\infty} \frac{e^{-ax} \sin x}{x} dx$$

$\therefore a=0$

$$\text{so, } \int_0^{\infty} \frac{e^{-ax} \sin x}{x} dx = \cot^{-1}(0) = \frac{\pi}{2}$$